

Avalanche-Enhanced Infrared Photodetection via Interlayer Absorption in a WSe₂/MoS₂ Heterostructure

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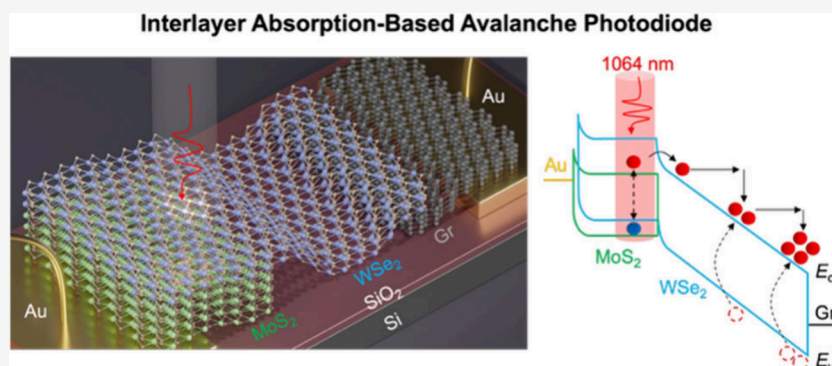
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ABSTRACT: Infrared photodetectors are crucial for a broad range of emerging optical applications. Type-II band alignment in two-dimensional (2D) heterostructures whose constituent layers possess visible-range bandgaps enables sub-bandgap infrared photoresponse via interlayer optical transitions. However, the responsivity of such devices remains limited by intrinsically weak optical absorption associated with interlayer transitions. Here, we demonstrate an interlayer-absorption avalanche photodiode based on a WSe₂/MoS₂ heterostructure, harnessing avalanche multiplication to overcome the weak interlayer absorption. Sub-bandgap infrared illumination (1,064 nm) generates photocurrent through interlayer absorption within the heterostructure, while avalanche multiplication is induced in an adjacent WSe₂ region. As a result, the device exhibits a 63-fold enhancement in responsivity, reaching 1 mA/W. These results establish avalanche multiplication as an effective internal-gain mechanism for interlayer-absorption photodetectors, highlighting a scalable route toward sensitive infrared detection using 2D semiconductor platforms.

KEYWORDS: infrared photodetection, interlayer absorption, 2D materials, van der Waals heterostructure, avalanche photodiodes

Infrared photodetection is essential for a variety of optical applications, including wearable optoelectronic devices, imaging sensors, and neuromorphic computing. A photodetector operates by converting incident photons into an electrical current, which requires materials with bandgaps smaller than the photon energy. Infrared photodetectors rely on materials capable of absorbing low-energy photons, motivating extensive studies of narrow-bandgap semiconductors, including III–V compounds, Hg_{1–x}Cd_xTe, and black phosphorus.^{1,2} However, these materials often face various significant challenges, such as large lattice mismatches with silicon (Si) substrates and complex and costly fabrication processes required to maintain material stability. Moreover, their inherent narrow bandgaps lead to high dark currents, which severely limit their sensitivity.

To address the limitations of conventional narrow-bandgap materials, two-dimensional (2D) semiconductors have recently emerged as promising platforms for infrared photodetection.^{3,4} In particular, van der Waals (vdW) heterostructures composed of transition metal dichalcogenides (TMDs) enable infrared

absorption through interlayer optical transitions, where electrons are excited from the valence band of one layer to the conduction band of another.^{5,6} Although individual TMD layers have relatively visible-range bandgaps, type-II band alignment in heterostructures such as WSe₂/MoSe₂,⁷ WS₂/MoS₂,⁸ and WSe₂/MoS₂⁹ facilitates low-energy interlayer absorption and thus photoresponse in the infrared range. However, the intrinsically thin absorption region and weak oscillator strength of interlayer transitions limit light–matter interaction, thereby constraining the overall responsivity of such devices.¹⁰

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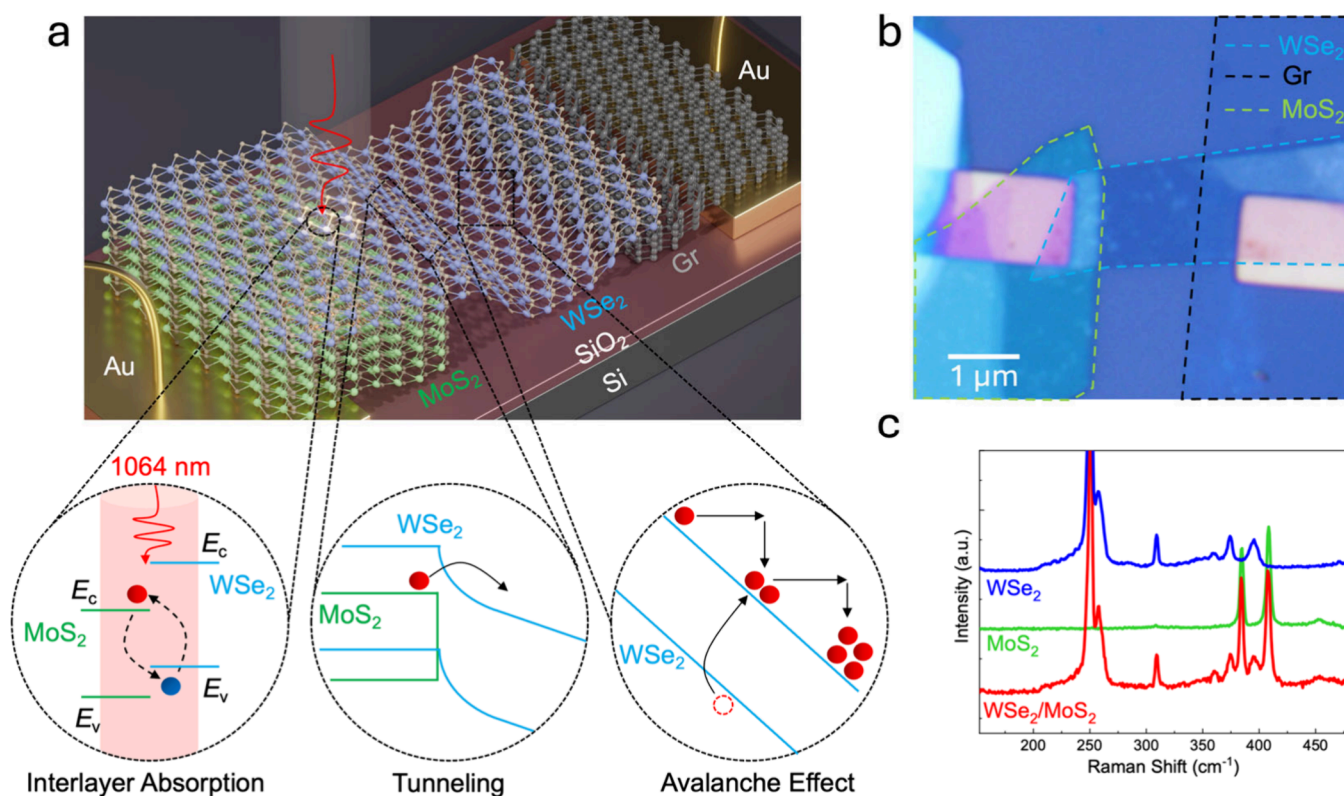


Figure 1. Device concept and characterizations. (a) The 3D schematic of the interlayer absorption-based avalanche photodiode with the WSe₂/MoSe₂ heterostructure (top). The band diagrams of the interlayer absorption in the heterostructure (bottom left), the tunneling at WSe₂/heterostructure (bottom middle), and the avalanche effect in WSe₂ (bottom right) are illustrated. (b) The optical microscopy image of the demonstrated 2D materials-based avalanche photodiode. (c) Raman spectra of WSe₂ (blue), MoSe₂ (green), and WSe₂/MoSe₂ heterostructure (red).

To further improve the limited optical absorption in interlayer photodetectors, several enhancement strategies have been explored. For example, plasmonic nanostructures such as Au nanopatterns can locally amplify the electromagnetic field through surface plasmon resonance (SPR), yielding more than a 20-fold increase in responsivity.¹¹ Other approaches include twist-angle engineering to maximize interlayer coupling,¹² electrostatic gating to accumulate interlayer excitons and enhance oscillator strength by over an order of magnitude,¹³ and hybrid integration with lead sulfide quantum dots (PbS QDs) to promote carrier transfer.¹⁴ While effective, these methods often suffer from wavelength selectivity, complex fabrication requirements, or the need for additional materials, which hinder scalability and broadband applicability. More importantly, these strategies primarily enhance absorption or carrier generation and do not introduce intrinsic gain, leaving the device responsivity fundamentally constrained by the weak interlayer absorption.

Here, we address this limitation by introducing avalanche multiplication as an intrinsic gain mechanism for interlayer-transition photodetectors, in which photoexcited carriers gain sufficient kinetic energy in a strong electric field to generate secondary electron–hole pairs through impact ionization. Although avalanche multiplication has been previously employed in conventional infrared photodiodes for internal gain, its implementation in interlayer-absorption-based 2D photodetectors remains largely unexplored because the interlayer charge transport often results in increased scattering due to momentum mismatch, while the effective narrow bandgap enhances tunneling within the interlayer region that

dominates over impact ionization, thereby suppressing stable avalanche multiplication. In our design, this challenge is addressed by spatially separating the absorption and multiplication processes within the device architecture. The WSe₂/MoSe₂ heterostructure functions as the absorption layer, enabling sub-bandgap infrared excitation via interlayer absorption. The photogenerated electrons are then efficiently injected into an adjacent WSe₂ multiplication region via field-assisted tunneling, where the strong electric field induces impact ionization and avalanche multiplication.^{15–17} This mechanism yields a 63-fold enhancement in responsivity, demonstrating that the avalanche effect can serve as an intrinsic gain pathway to overcome the weak optical absorption of interlayer transitions. The proposed approach thus establishes a new route for high-sensitivity infrared detection using well-developed 2D TMD materials with visible-range bandgaps.

The three-dimensional (3D) schematic of the prototypical interlayer absorption avalanche photodiode is illustrated in Figure 1a. It consists of the heterostructure (WSe₂/MoSe₂-overlapped region) and the WSe₂ layer (nonoverlapped region). The key operating principle is that sub-bandgap infrared photons, whose energies are lower than the intrinsic bandgaps of both WSe₂ and MoSe₂, are absorbed in the WSe₂/MoSe₂ heterostructure (the overlapped region) through interlayer absorption (bottom left in Figure 1a).¹⁸ The estimated interlayer bandgap is 0.82 eV with the conduction band offset of ~0.4 eV in the heterostructure, allowing sub-bandgap infrared photodetection with a 1,064 nm laser (see Supporting Figure S2). The resulting photoexcited electrons

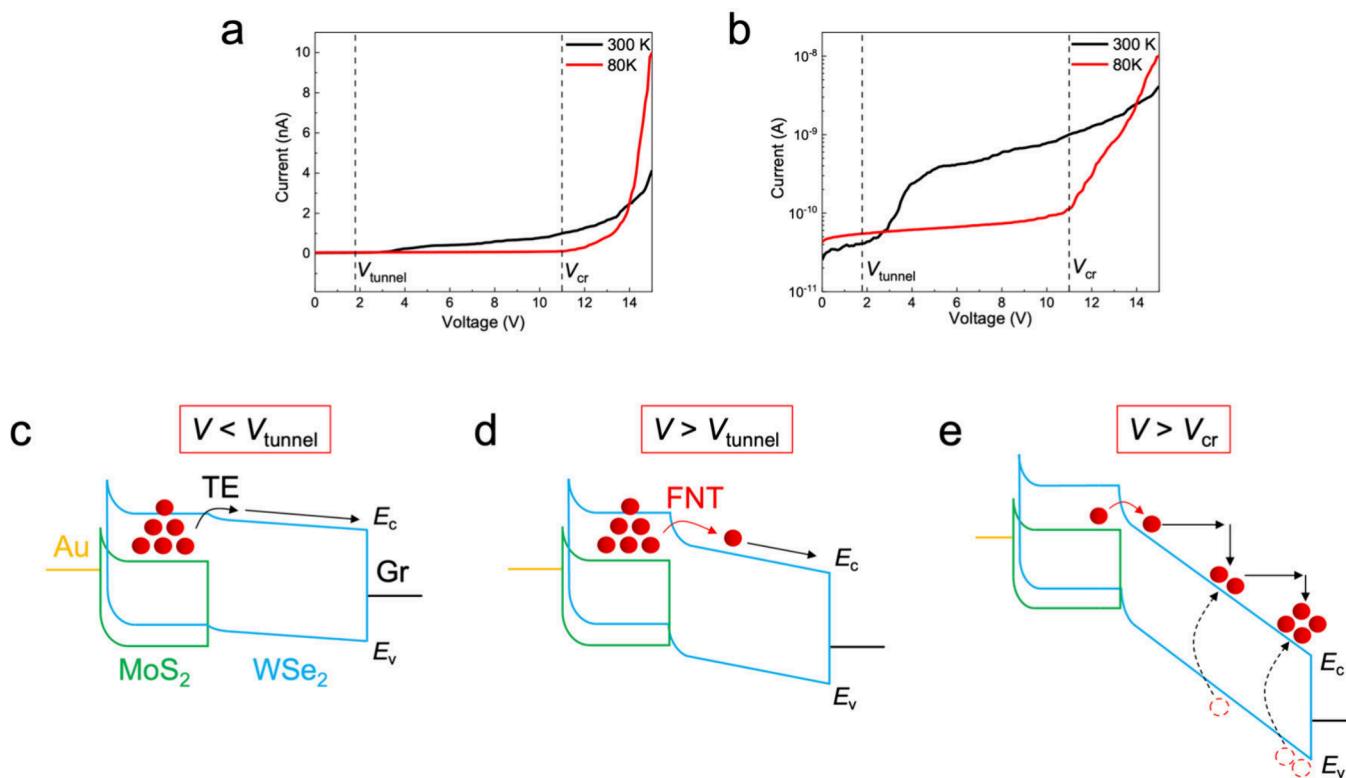


Figure 2. Electrical properties of the interlayer absorption avalanche photodiode. $I_{\text{dark}}-V$ curves ($V_g = 40$ V) of the device at two different temperatures, 80 and 300 K in (a) linear- and (b) log-scales. The band diagram of the device at (c) $V < V_{\text{tunnel}}$, (d) $V > V_{\text{tunnel}}$, and (e) $V > V_{\text{cr}}$ under the avalanche mode.

are then injected into the adjacent WSe_2 region via efficient tunneling (bottom middle in Figure 1a).¹⁹ There, a strong electric field triggers avalanche multiplication, leading to substantial photocurrent amplification and enhanced infrared photoresponse (bottom right in Figure 1a).¹⁶

The optical microscopy image of the device is shown in Figure 1b. The Au-patterned SiO_2 substrate was prepared, and then the 2D vdW heterostructure consisting of MoS_2 , WSe_2 , and graphene was assembled on the substrate using a dry-transfer method (see Supporting Figure S1). The vdW contact between MoS_2 and Au forms a Schottky barrier that suppresses unwanted electron injection from Au into MoS_2 .²⁰ In contrast, the WSe_2 /graphene interface provides low contact resistance,²¹ enabling the accumulation of a strong electric field in the WSe_2 region required for avalanche multiplication.^{22–25} In addition, WSe_2 is stacked on top of MoS_2 in the heterostructure (Supporting Figure S1), which improves the modulation efficiency of the charge transfer.^{19,25–27}

Raman spectroscopy was conducted to characterize the exfoliated MoS_2 and WSe_2 layers (Figure 1c). The Raman spectrum of WSe_2 (blue curve) shows a peak at 308 cm^{-1} , indicating multilayer WSe_2 .²⁸ The Raman spectrum of MoS_2 (green curve) exhibits two characteristic peaks near 400 cm^{-1} with a separation larger than 20 cm^{-1} , which is consistent with multilayer MoS_2 .²⁹ The Raman spectrum from the WSe_2 / MoS_2 overlapping region (red curve) shows peaks from both multilayer WSe_2 and MoS_2 .³⁰

To characterize the electrical properties of the device in the dark, the dark current–voltage ($I_{\text{dark}}-V$) characteristics were measured at different temperatures (80 and 300 K) and displayed in both linear (Figure 2a) and logarithmic scales (Figure 2b). The back-gate voltage (V_g) was set to 40 V unless

otherwise stated. At high V_g , the electric field is concentrated in the adjacent WSe_2 region, increasing the tunneling efficiency. The typical tunneling transmission probability can be expressed by the following equation:¹⁹

$$T(F) = \exp\left(-\frac{\pi(m_{\text{tunnel}})^{1/2}\Delta E_g^{3/2}}{2\sqrt{2}\hbar qF}\right) \quad (1)$$

where m_{tunnel} is the carrier effective mass in the tunneling direction, E_g is the effective bandgap, q is the electron charge, $\hbar$ is the reduced Planck constant, and F is the electric field across the semiconductor. Increasing V_g induced a high n -doping density in the heterostructure. The heavy n -doped heterostructure is less band-bended under the applied voltage, while the strong electric field is concentrated into the adjacent WSe_2 region. As a result, the enhanced tunneling transmission probability enables efficient tunneling-assisted electron injection from the heterostructure into WSe_2 (Supporting Figure S3a–f).^{31,32}

As shown in Figure 2a and 2b, the dark current exhibits noticeable changes at two characteristic voltages: the tunneling voltage (V_{tunnel}) and the critical voltage (V_{cr}). At low forward bias ($V < V_{\text{tunnel}}$), a wide and high potential barrier forms in the WSe_2 region (the nonoverlapping region). In this regime, electron transport is dominated by thermionic emission (TE) over the barrier. However, due to the large barrier height in WSe_2 , the current remains negligible even at 300 K, indicating that the thermal excitation alone is insufficient to inject carriers from the heterostructure into WSe_2 (Figure 2c).³³

As the applied bias increases beyond V_{tunnel} , the $I_{\text{dark}}-V$ curves exhibit clear temperature dependence. In this regime, the dark current increases at 300 K but remains nearly

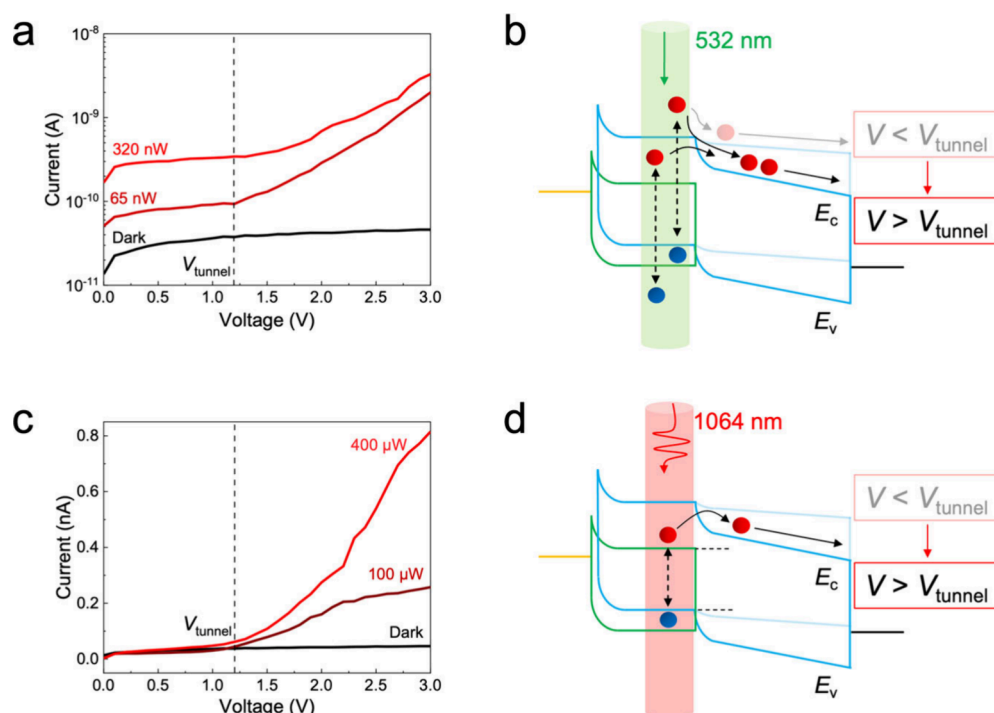


Figure 3. Photoresponse in forward bias from 0 to 3 V. $I_{\text{photo}}-V$ curves for the device under (a) 532 nm and (c) 1064 nm illuminations. The band diagrams of the device at different forward bias ($V < V_{\text{tunnel}}$ and $V > V_{\text{tunnel}}$) under (b) 532 nm and (d) 1064 nm illuminations.

unchanged at 80 K. With increasing bias, the electric field bends the WSe_2 barrier into a triangular profile, enabling Fowler-Nordheim tunneling (FNT).³⁴ At 300 K, thermally excited electrons tunnel through the triangular WSe_2 barrier and are injected into the WSe_2 region (Figure 2d). At low temperature (80 K), however, the reduced thermal population limits the tunneling current despite the presence of the triangular barrier. The transition from TE to FNT is confirmed by the Fowler-Nordheim plot of $\ln(I_{\text{dark}}/V^2)$ versus $1/V$ at 300 K (Supporting Figure S3g).

At higher forward bias beyond V_{cr} , the dark current increases exponentially at both temperatures (Figure 2b), indicating the onset of avalanche multiplication. The onset voltage shifts from 14 V at 300 K to 11 V at 80 K. Such a reduction in the onset voltage with decreasing temperature has been reported as a characteristic feature of avalanche processes,^{15,17} distinguishing this regime from the TE and FNT regime.³¹ Further evidence of avalanche multiplication is provided by the extracted ionization index $n = 12.7$ (Supporting Figure S3h), which is consistent with previously reported values.^{35,36} This carrier multiplication results in the pronounced increase in dark current observed in Figure 2b. The entire process can be understood as follows: under high forward bias, the strong electric field across the WSe_2 region accelerates carriers injected from the heterostructure via FNT. These energetic carriers undergo impact ionization in WSe_2 , generating secondary electron-hole pairs and initiating avalanche multiplication (Figure 2e).

To investigate the initial photoresponse of the device, the photocurrent-voltage ($I_{\text{photo}}-V$) characteristics were measured under 532 nm and 1,064 nm illumination (Figure 3a and 3c). The corresponding band diagrams are also shown in Figure 3b and 3d, respectively. The 532 nm excitation induces intralayer absorption in both WSe_2 and MoS_2 , whereas the 1,064 nm illumination predominantly excites interlayer

transitions within the heterostructure. For both illumination wavelengths, a distinct change in photocurrent behavior is observed near the V_{tunnel} .

Under 532 nm illumination, both WSe_2 and MoS_2 layers absorb photons through intralayer transitions. As a result, the photocurrent increases with increasing optical power even at biases below V_{tunnel} (Figure 3a). Such a photocurrent increase can be understood from the band diagram shown in Figure 3b. Photoexcited electrons in WSe_2 can be readily transferred into the nonoverlapped WSe_2 region, generating the photocurrent. In contrast, photocarriers generated in the MoS_2 layer are largely blocked by the potential barrier in WSe_2 when the bias is below V_{tunnel} , preventing their efficient transport into the WSe_2 region. Once the forward bias exceeds V_{tunnel} , the photocurrent increases more rapidly. In this regime, the strong electric field bends the conduction and valence bands of WSe_2 and transforms the barrier into a triangular profile, allowing the photocarriers in MoS_2 to tunnel into the WSe_2 conduction band via FNT, which leads to a substantial increase in photocurrent.³⁷

We then applied 1,064 nm illumination to selectively induce interlayer absorption in the $\text{WSe}_2/\text{MoS}_2$ heterostructure. At biases below V_{tunnel} , the photocurrent remains negligible, indicating that the electrons generated via the interlayer absorption cannot overcome the large potential barrier in WSe_2 (Figure 3c). Once the forward bias exceeds V_{tunnel} , efficient tunneling of interlayer-excited electrons is activated, and the photocurrent begins to emerge, similar to the behavior observed under intralayer excitation (Figure 3a). The corresponding band diagram in Figure 3d illustrates this process: interlayer photoexcitation generates electrons within the heterostructure, and these carriers subsequently tunnel through the triangular WSe_2 barrier into the WSe_2 conduction band, marking the onset of efficient carrier injection from the heterostructure into the WSe_2 region. The repeatability of the

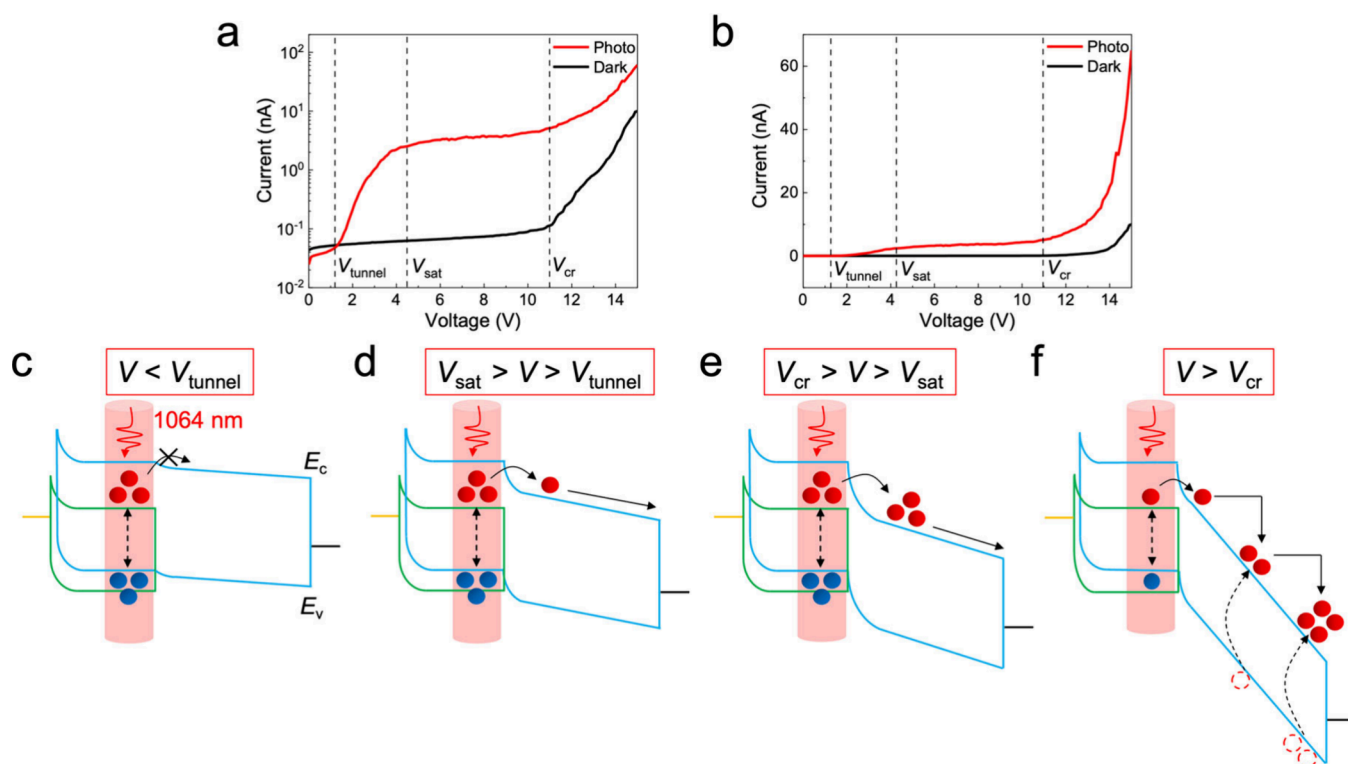


Figure 4. Photoresponse in forward bias from 0 to 15 V. I – V curves in dark and 1064 nm illumination in (a) log- and (b) linear-scales. The dark current from Figure 2a and b is included for reference. The band diagrams of the device (c) $V < V_{\text{tunnel}}$, (d) $V_{\text{sat}} > V > V_{\text{tunnel}}$, (e) $V_{\text{cr}} > V > V_{\text{sat}}$, and (f) $V > V_{\text{cr}}$.

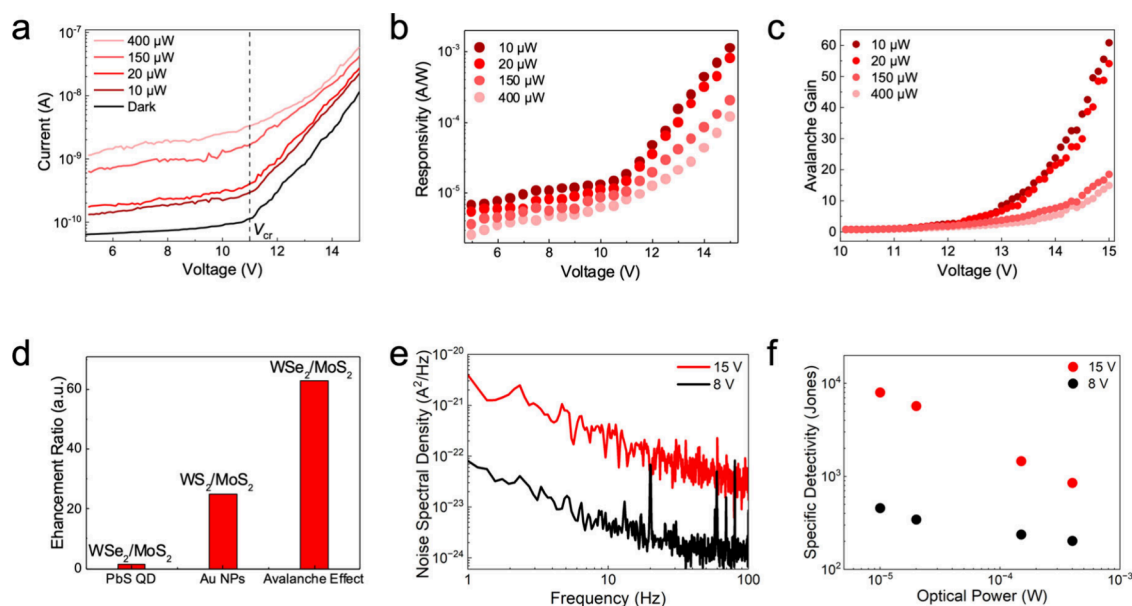


Figure 5. Avalanche photodetection performance of the WSe₂/MoS₂ heterostructure photodiode. (a) I_{photo} – V curves at different optical powers. (b) The extracted avalanche gain as a function of forward bias. (c) The responsivity for the device before and after the avalanche mode. (d) The comparison of the enhancement ratio with the reported interlayer absorption-based photodetectors from references: PbS QD-enhanced interlayer absorption photodetectors,¹⁴ Au nanopatterned interlayer absorption photodetectors.¹¹ (e) Noise spectral density measured at 8 and 15 V under 80 K. (f) Extracted specific detectivity (D^*) as a function of optical powers at 8 and 15 V.

photoresponse was confirmed by the time-dependent measurements shown in Figure S4a and S4b. Supporting Figure S4c and S4d show negligible photocurrent at 1,550 nm, indicating that the photon energy of 0.80 eV is below the interlayer bandgap. Combined with the estimated type-II band alignment in Figure S2, these results are consistent with an interlayer

bandgap of ~ 0.82 eV, corresponding to a cutoff wavelength near 1,500 nm. As the forward bias increases further, the injected carriers experience strong electric fields in the WSe₂ region, eventually leading to avalanche multiplication, as discussed in the following section.

To investigate the photoresponse near $V = V_{cr}$, the forward bias was further increased up to 15 V under 1,064 nm illumination. The current–voltage (I – V) characteristics measured at 80 K are shown in Figure 4a and 4b in logarithmic and linear scales, respectively. Notable changes in the photocurrent are observed at three distinct voltages: V_{tunnel} , the saturation voltage (V_{sat}), and V_{cr} .

At $V < V_{tunnel}$, as previously mentioned, carrier injection from the heterostructure into WSe_2 is suppressed by the large potential barrier, leading to negligible photocurrent (Figure 4c). Once the applied bias exceeds V_{tunnel} , the FNT becomes activated, allowing photoexcited electrons generated in the WSe_2/MoS_2 heterostructure to tunnel through the triangular barrier into the WSe_2 region. As a result, the photocurrent increases significantly (Figure 4d).

As the forward bias reaches the second voltage point, V_{sat} , the photocurrent begins to saturate. This saturation indicates that the majority of the photoexcited carriers in the heterostructure are fully injected into WSe_2 (Figure 4e),³⁸ such that the further increases in the forward bias no longer significantly enhance the photocurrent.

Lastly, beyond the third voltage point, V_{cr} , the photocurrent exhibits an exponential rise. In this regime, the strong electric field in the WSe_2 region steepens its band bending and accelerates the injected electrons, enabling impact ionization and the generation of secondary electron–hole pairs. This process triggers the avalanche multiplication process, resulting in substantial internal gain in the photocurrent. The corresponding band diagram in Figure 4f illustrates this process, where the steeply tilted WSe_2 conduction band enables carrier acceleration and repeated impact ionization.

To evaluate the device performance in the avalanche regime, the photocurrent–voltage (I_{photo} – V) characteristics were measured near V_{cr} under various optical powers (Figure 5a). As discussed previously in Figure 4f, once the forward bias exceeds V_{cr} , impact ionization is activated in the WSe_2 multiplication region, leading to rapid carrier multiplication and a sharp increase in photocurrent. Based on the I_{photo} – V curves in Figure 5a, the avalanche gain (G) was extracted using the following equation:³⁹

$$G = \frac{I_{ph} - I_{ph0}}{I_{dark} - I_{dark0}} \quad (2)$$

where I_{ph0} and I_{dark0} are the photocurrent and dark current at $V = V_{cr}$, respectively. As shown in Figure 5b, the avalanche gain increases with forward bias, reaching 15 at 15 V at a high optical power of 400 μ W and up to 63 at a low optical power of 10 μ W. The inverse dependence of avalanche gain on optical power is attributed to space-charge screening and heating effects.^{40,41} At high optical powers, accumulated carriers partially screen the applied electric field, suppressing the effective impact ionization. In addition, high optical power heats up around the illumination region. The increased temperature enhances carrier scattering, reducing the impact ionization. Similar behavior has also been widely observed in bulk and 2D materials-based avalanche photodetectors.^{3,42–44}

To further assess the performance, the responsivity (R) was obtained using the following equation:⁴⁵

$$R = \frac{I_{ph} - I_{dark}}{P_{opt}} \quad (3)$$

where P_{opt} is the injected optical power. As shown in Figure 5c, the responsivity increases moderately with forward bias before the onset of avalanche multiplication at V_{cr} (up to $\sim 17 \mu$ A/W at 11 V), and rises sharply at biases higher than V_{cr} , reaching 100 μ A/W at 400 μ W and 1 mA/W at 10 μ W under 15 V bias.

To benchmark the performance of our interlayer absorption-based avalanche photodiode, we compare the responsivity enhancement achieved in this work with previously reported interlayer absorption-based photodetectors (Figure 5d). Previous studies improved interlayer photodetection by introducing additional optical structures or materials. For example, PbS QDs integrated on WSe_2/MoS_2 heterostructures improved responsivity by $1.8\times$ (160 μ A/W),¹⁴ while plasmonic Au nanostructures on WSe_2/MoS_2 heterostructures achieved $25\times$ enhancement through improved optical confinement.¹¹ In contrast, the present device demonstrates a $63\times$ enhancement solely through avalanche multiplication in WSe_2 , without requiring additional nanostructures or external absorbers.

To analyze the noise characteristics of the device, the noise spectral density was measured at 80 K under $V_g = 40$ V in the dark. Figure 5e shows the noise spectral density in the frequency range from 1 to 100 Hz. The current-noise spectral densities were measured to be 6.03×10^{-23} A²/Hz at 8 V and 2.20×10^{-21} A²/Hz at 15 V under 1 Hz. The enhanced avalanche gain induces high shot noise for the device, leading to the high noise spectral density at 15 V. The inverse frequency dependence indicates that $1/f$ noise dominates within the measured bandwidth. The noise equivalent power (NEP) can be calculated as $\sqrt{S}/R$, where S is the current-noise spectral density. The NEP decreases from 7.2×10^{-7} W/Hz^{1/2} at 8 V to 4.2×10^{-8} W/Hz^{1/2} at 15 V under the optical power of 10 μ W. Despite the increased noise density at 15 V, avalanche multiplication substantially reduced the NEP owing to the higher responsivity.

The specific detectivity (D^*) was extracted using the following equation:

$$D^* = \frac{R\sqrt{A \times \Delta f}}{\sqrt{\langle i_n^2 \rangle}} \quad (4)$$

where A is the area of the heterostructure, Δf is the bandwidth, and i_n is the noise current in the dark. Integrating the measured noise spectral density over the experimental bandwidth in Figure 5e yields a mean-square noise current, $\langle i_n^2 \rangle$, that increases from 5.50×10^{-22} A² at 8 V to 1.94×10^{-20} A² at 15 V over the measured bandwidth range. Under the optical power of 10 μ W, the extracted D^* increases from 4.56×10^2 Jones at 8 V to 7.99×10^3 Jones at 15 V owing to the responsivity enhancement enabled by avalanche multiplication (Figure 5f). The strong optical-power dependence of D^* at 15 V originates from the high avalanche gain at low optical powers. Although avalanche multiplication improved D^* , the absolute detectivity remains limited by the intrinsically weak interlayer absorption.

In summary, we demonstrated sensitive infrared photodetection in interlayer absorption-based 2D heterostructures by exploiting avalanche multiplication as an intrinsic gain mechanism. By utilizing a 1,064 nm wavelength—below the intrinsic bandgaps of both WSe_2 and MoS_2 —we successfully induced interlayer photoexcitation. The FNT enabled efficient carrier injection from the heterostructure into WSe_2 , as confirmed by the bias-dependent photoresponse. Further

increase in bias triggered avalanche effect in WSe_2 , resulting in a large internal gain of 63, as well as responsivity reaching 1 mA/W under low optical power. Unlike previously reported enhancement methods that rely on external nanostructures and additional materials, our approach achieves substantial internal gain through bias-controlled carrier multiplication. This finding opens up new possibilities for compact, robust, and highly sensitive infrared photodetectors based on well-developed 2D materials.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.6c01893>.

- (1) Device fabrication; (2) simulated band diagrams; (3) electrical properties of the device; (4) photocurrent characteristics of the device (PDF)

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Notes

The authors declare no competing financial interest.

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